

ECE 343 Project 6:

Computation of Expectations and Autocorrelation

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1 Introduction

In this project we analyze the random process

$$X(t) = \cos(\omega t + \Theta),$$

where Θ is uniform on $[0, 2\pi)$. We will compute the mean and the autocorrelation function.

2 Expectation of $\cos(\omega t + \Theta)$

Let $\Theta \sim \text{Uniform}(0, 2\pi)$, so its pdf is

$$f_{\Theta}(\theta) = \frac{1}{2\pi}, \quad 0 \leq \theta < 2\pi.$$

Then

$$E[\cos(\omega t + \Theta)] = \int_0^{2\pi} \cos(\omega t + \theta) \frac{d\theta}{2\pi} = \frac{1}{2\pi} \int_{\omega t}^{\omega t + 2\pi} \cos(u) du = \frac{\sin(u)}{2\pi} \Big|_{u=\omega t}^{\omega t + 2\pi} = 0.$$

3 Autocorrelation $R_{XX}(t_1, t_2)$

By definition,

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = E[\cos(\omega t_1 + \Theta) \cos(\omega t_2 + \Theta)] = \frac{1}{2\pi} \int_0^{2\pi} \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta) d\theta.$$

Use $\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$ to get

$$\begin{aligned} R_{XX}(t_1, t_2) &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} [\cos(\omega(t_1 + t_2) + 2\theta) + \cos(\omega(t_1 - t_2))] d\theta \\ &= \underbrace{\frac{1}{4\pi} \int_0^{2\pi} \cos(\omega(t_1 + t_2) + 2\theta) d\theta}_{=0} + \frac{1}{4\pi} \int_0^{2\pi} \cos(\omega(t_1 - t_2)) d\theta \\ &= 0 + \frac{2\pi}{4\pi} \cos(\omega(t_1 - t_2)) = \frac{1}{2} \cos(\omega(t_1 - t_2)). \end{aligned}$$

The first integral is equivalent to zero because it has a period of 2π and is integrated over its full period, which will have equal parts positive and negative. If you carry out the calculation it will simplify down to the same sine term added and subtracted with itself, cancelling out. Since the autocorrelation function depends only on $\tau = t_1 - t_2$, the process is wide-sense stationary because it can be written as a function of the difference between any two points:

$$R_{XX}(\tau) = \frac{1}{2} \cos(\omega\tau).$$

4 Wiener–Khinchin Theorem

The Wiener–Khinchin theorem states that, for a wide-sense stationary process $X(t)$, the power spectral density (PSD) $S_{XX}(\omega)$ is the Fourier transform of its autocorrelation $R_{XX}(\tau)$:

$$S_{XX}(\omega') = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega'\tau} d\tau.$$

Since from Section 2,

$$R_{XX}(\tau) = \frac{1}{2} \cos(\omega\tau),$$

we have

$$S_{XX}(\omega') = \int_{-\infty}^{\infty} \frac{1}{2} \cos(\omega\tau) e^{-j\omega'\tau} d\tau.$$

Use the identity

$$\cos(\omega\tau) = \frac{1}{2} (e^{j\omega\tau} + e^{-j\omega\tau})$$

to get

$$\begin{aligned} S_{XX}(\omega') &= \frac{1}{2} \int_{-\infty}^{\infty} \frac{1}{2} (e^{j\omega\tau} + e^{-j\omega\tau}) e^{-j\omega'\tau} d\tau \\ &= \frac{1}{4} \int_{-\infty}^{\infty} (e^{j(\omega-\omega')\tau} + e^{-j(\omega+\omega')\tau}) d\tau. \end{aligned}$$

Recall the Fourier transform of a complex exponential is a delta:

$$\int_{-\infty}^{\infty} e^{j(\alpha)\tau} d\tau = 2\pi \delta(\alpha).$$

Applying that here,

$$\int_{-\infty}^{\infty} e^{j(\omega-\omega')\tau} d\tau = 2\pi \delta(\omega - \omega'), \quad \int_{-\infty}^{\infty} e^{-j(\omega+\omega')\tau} d\tau = 2\pi \delta(\omega + \omega').$$

Therefore

$$S_{XX}(\omega') = \frac{1}{4} \left[2\pi \delta(\omega - \omega') + 2\pi \delta(\omega + \omega') \right] = \frac{\pi}{2} [\delta(\omega' - \omega) + \delta(\omega' + \omega)].$$

That is, the spectrum consists of two lines at $\omega' = \pm\omega$, each of weight $\pi/2$.

MATLAB Results

Ideal Power Spectral Density

The section above showed that for the function $\text{Acos}(\omega t + \theta)$, with θ a uniform random variable between 0 and 2π , the ideal power spectrum density is two impulses. It also showed that the power spectral density of a function is the Fourier Transform of the autocorrelation. The Fourier Transform is a linear operator, and therefore the power spectral density of our function can be made from a linear superposition of three sinusoidal terms, all contributing two scaled impulses.

There is an additional term coinciding with the error noise of our random process. This white gaussian noise has a constant PSD for all frequencies. Combine that with the linear combination of PSD's for sinusoids, we arrive with the following equation for representing our power spectral density (PSD):

$$\frac{\pi}{2} \sum_{i=1}^3 (A_i [\delta(\omega - \omega_i) + \delta(\omega + \omega_i)]) + \sigma_e^2$$

The graph of the PSD of our signal therefore is three peaks (impulses) and a noise floor which is constant for all frequencies.

We now want to examine the computation of this PSD using the Averaging of the Periodogram method, by analyzing how a change in the characteristic values Q, L, K, and R will affect resolution and noise tolerance.

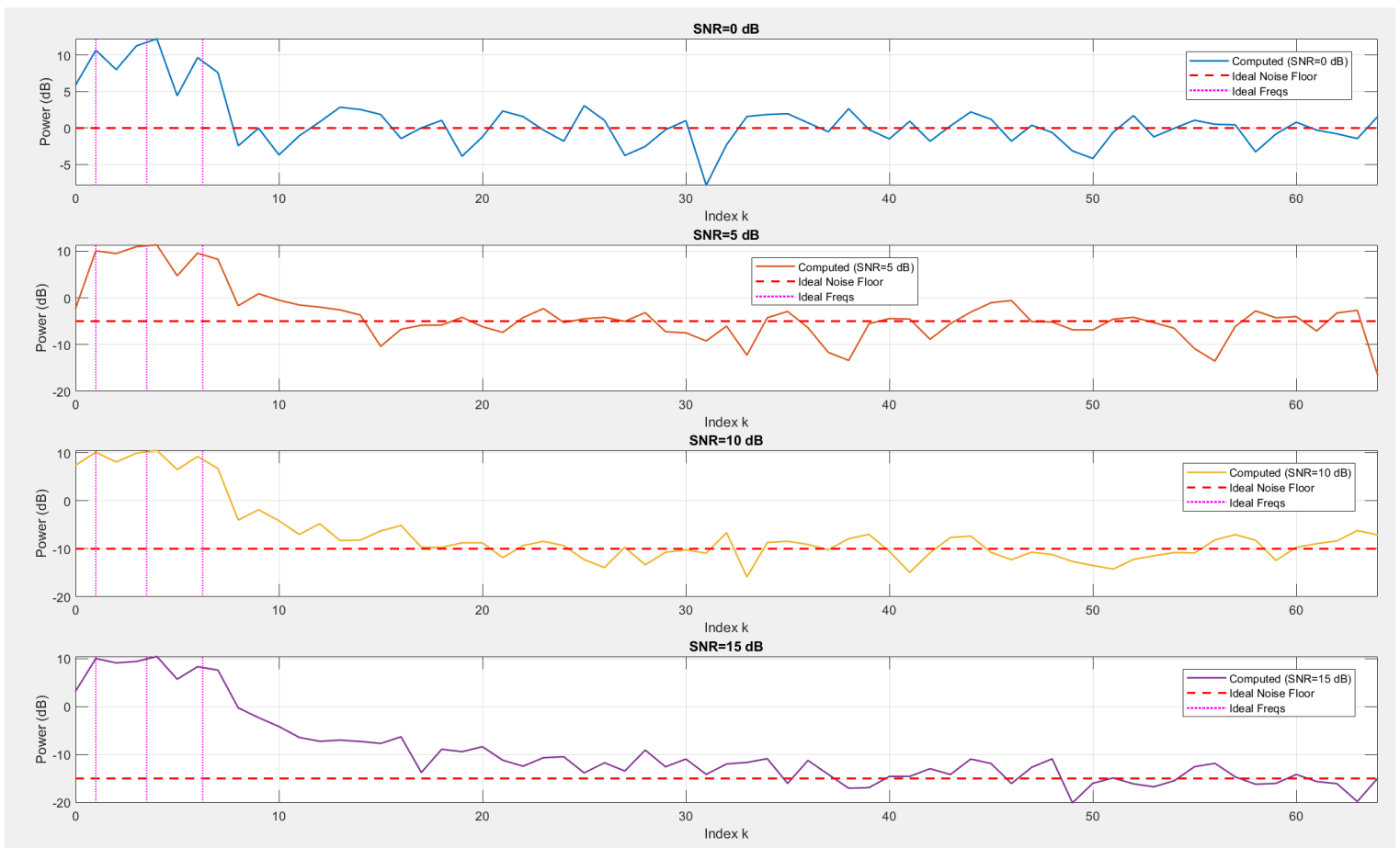
This analysis will be broken down into the following sections, where the method for computation defines the R value (either $R=L$ or $R=L/2$), and K is defined by Q and L.

- **Bartlett Method: Rectangular Window** with No Overlap ($R=L$)
 - Show computation for four SNR levels with L, Q, K, and R constant.
 - Show the computation for two SNR levels as L changes and Q remains constant.
 - Show the computation for two SNR levels as Q changes and L remains constant.
- **Welch Method: Rectangular Window** with Overlap ($R=L/2$)
- **Welch Method: Kaiser Window** with Overlap ($R=L/2$)

Bartlett Method: Rectangular Window

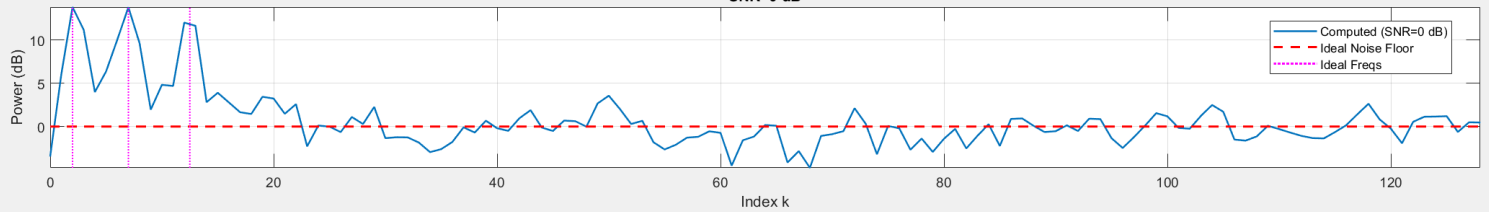
First, I will show two examples of constant Q , L , K , and R , but changing SNR values for three different combinations of those parameters.

$Q = 256$, $L = 64$

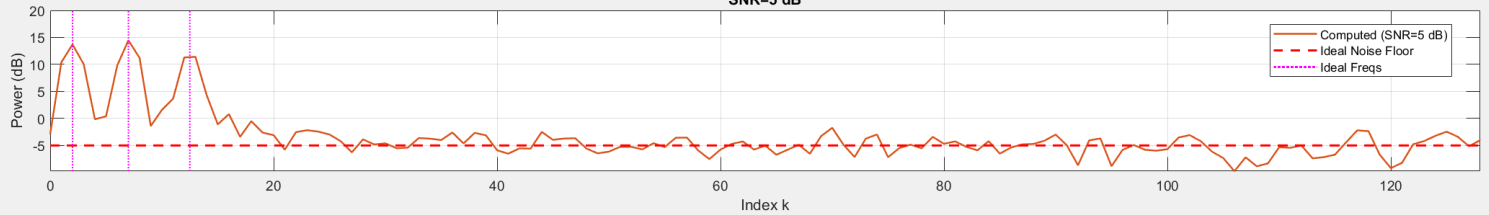


Q=1024, L=128

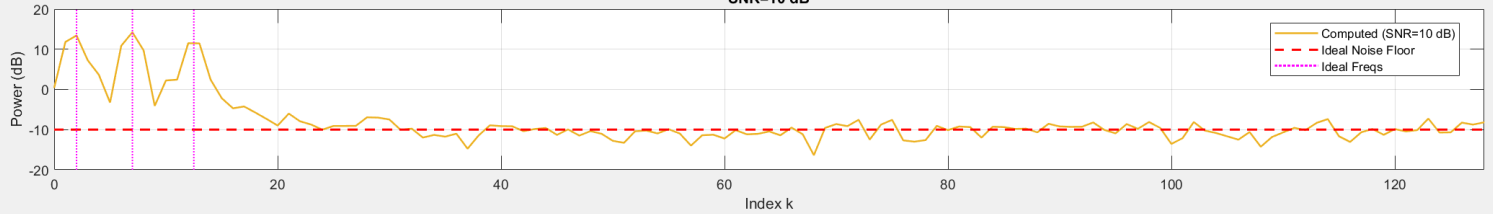
SNR=0 dB



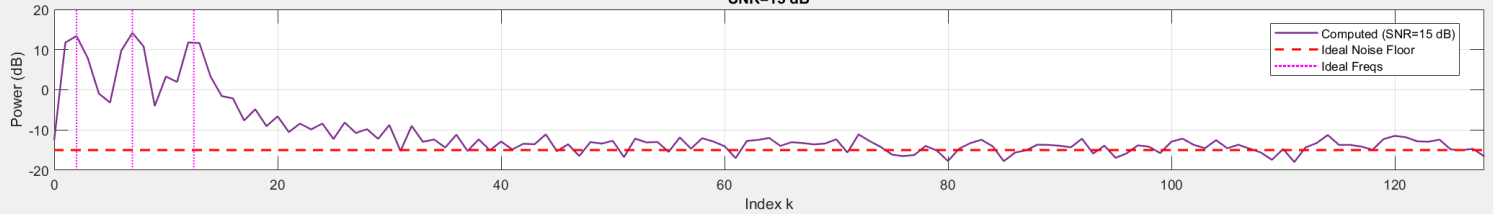
SNR=5 dB



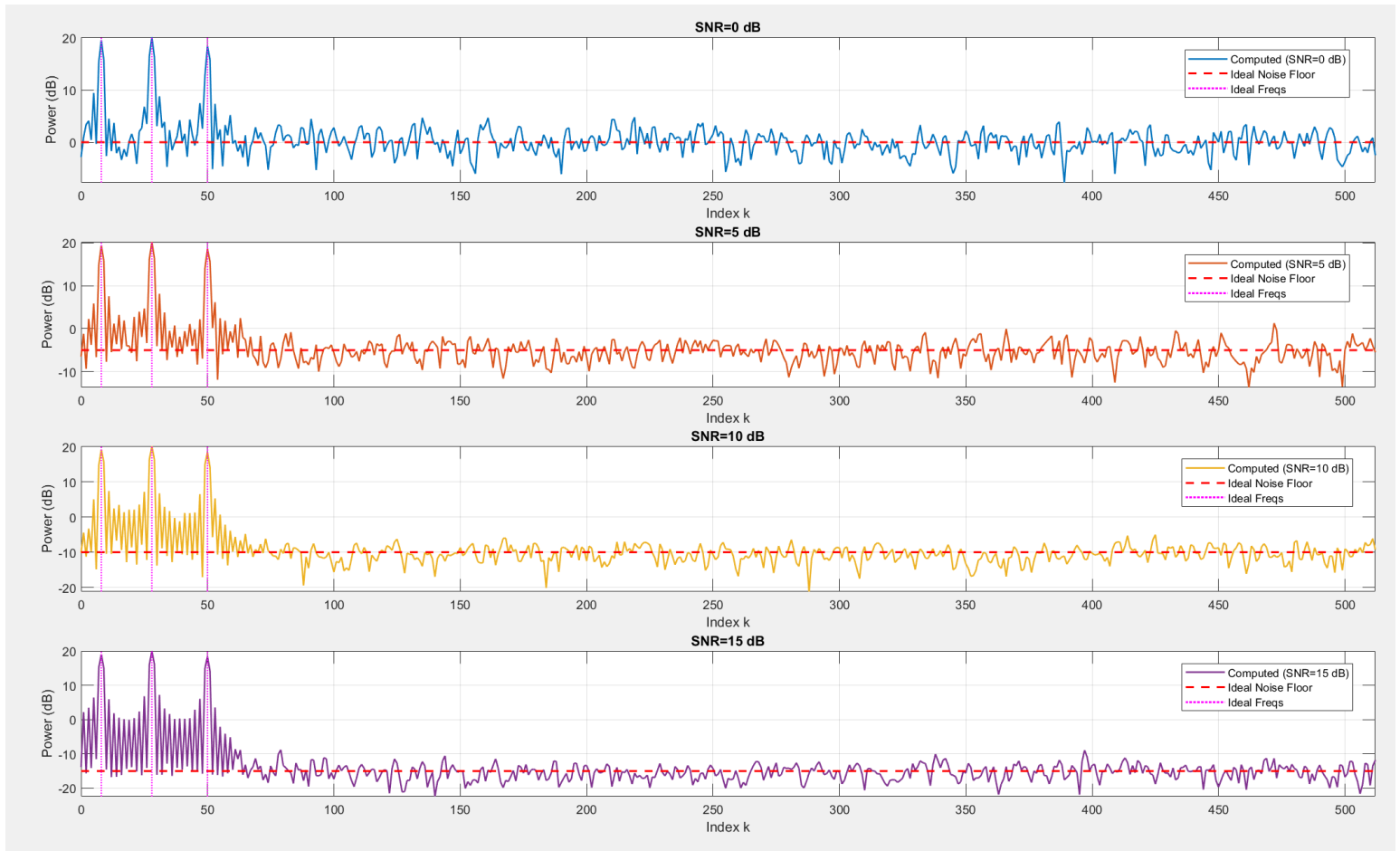
SNR=10 dB



SNR=15 dB



Q=2048, L=512

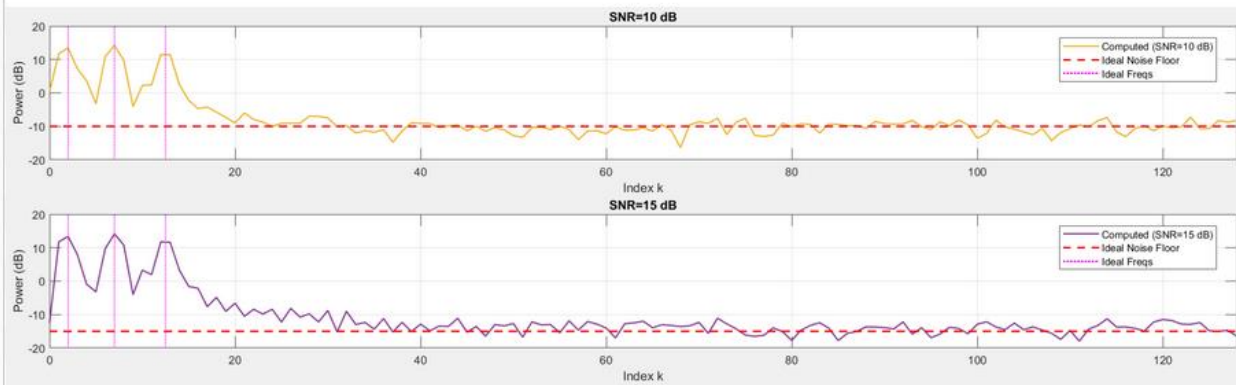


From these figures we can note for a given set of values, as SNR increases, the noise floor shifts down, however, the actual deviation caused by the noise does not change a significant amount.

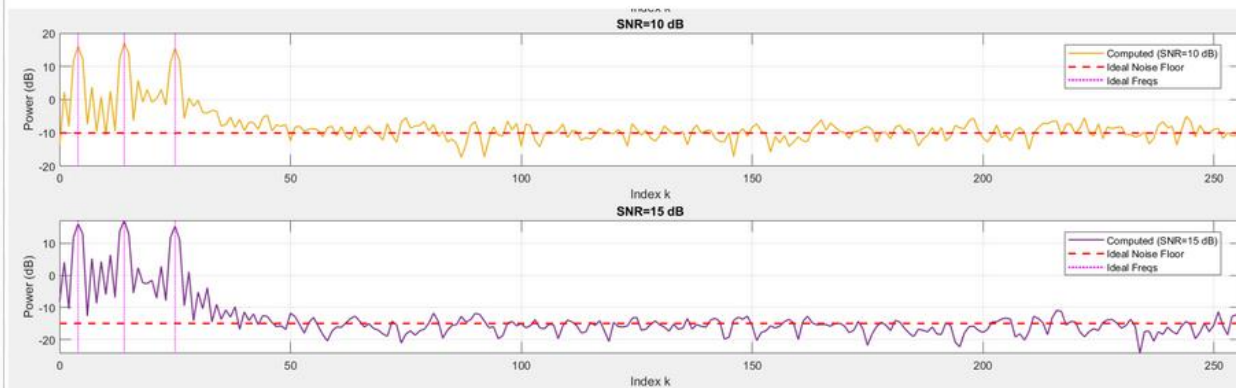
But, we cannot tell how each individual parameter like L and K affect the resolution and noise tolerance. To do this, I will vary one parameter while keeping the rest the same.

Changing L vs. Constant Q and SNR

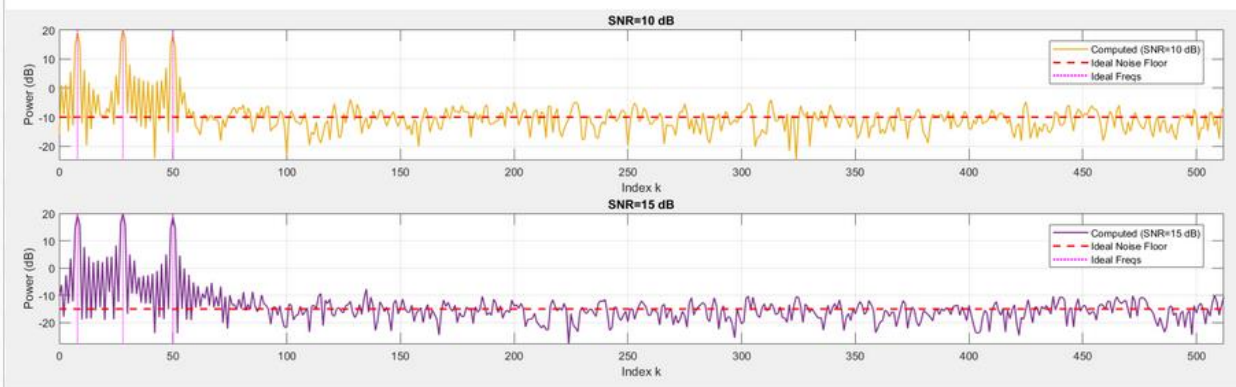
Q = 1024, L=128



Q = 1024, L=256



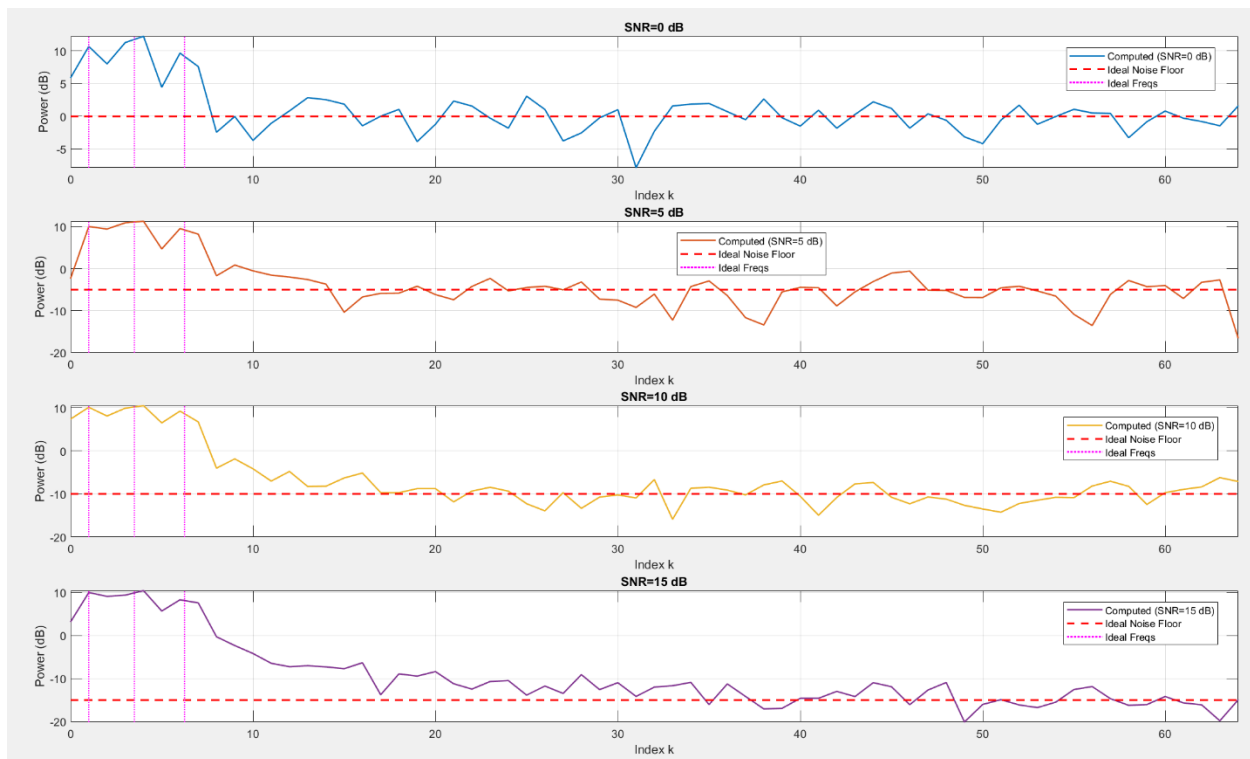
Q = 1024, L=512



As we change L, with all else held constant, we can see that the sampling rate (in the frequency domain) increases, and also that the amount of noise decreases with smaller L values. A smaller L means that there will be more subdivisions (since $R=L$ as well), and the final step of the periodogram averaging is to average over all those subdivisions. With more

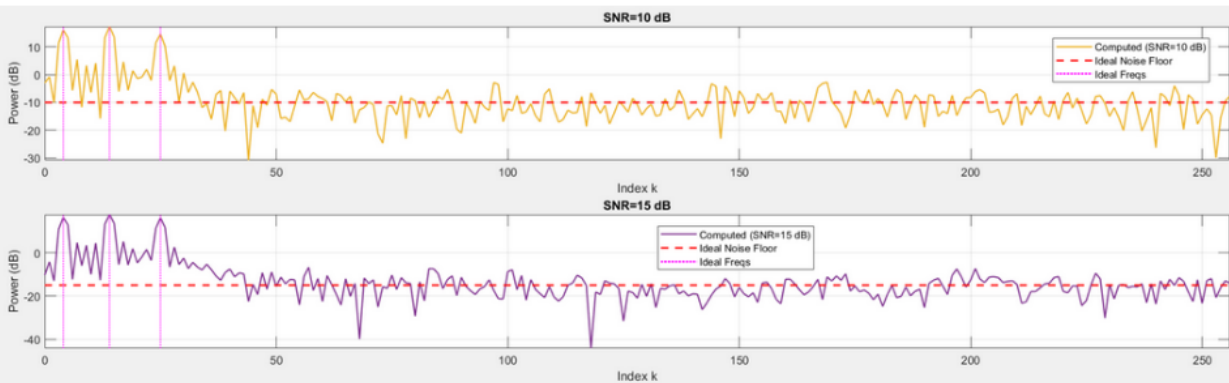
values to average, the graph will be smoother and less noisy, so this correlation makes sense.

We can also see that the resolution decreases as L decreases. This phenomenon eventually will reach the point where we cannot distinguish the three peaks at all, despite knowing what the frequencies are beforehand. In practice, even with $L=128$, we could not confidently say that the PSD represents a sum of sines with some white noise. The figure below shows that even with zero noise, if the window length is not high enough, we cannot properly do the FFT to find the PSD.

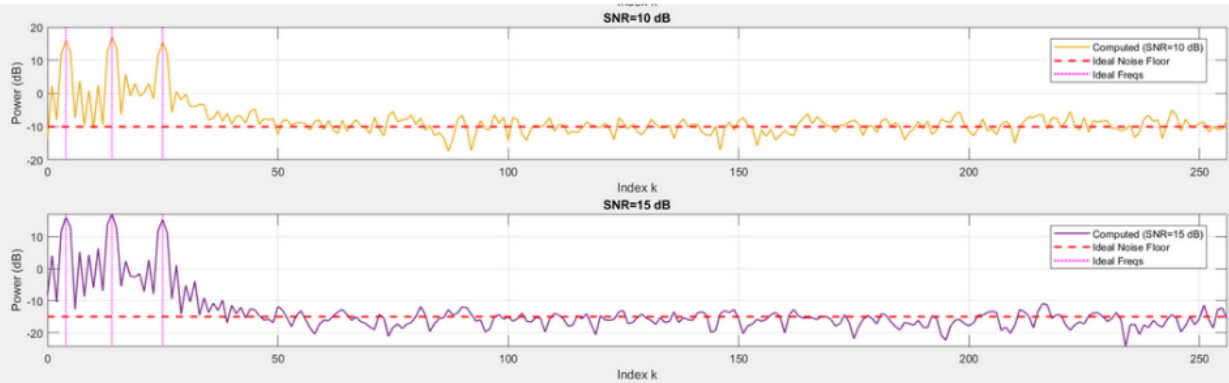


Changing Q vs. Constant L and SNR

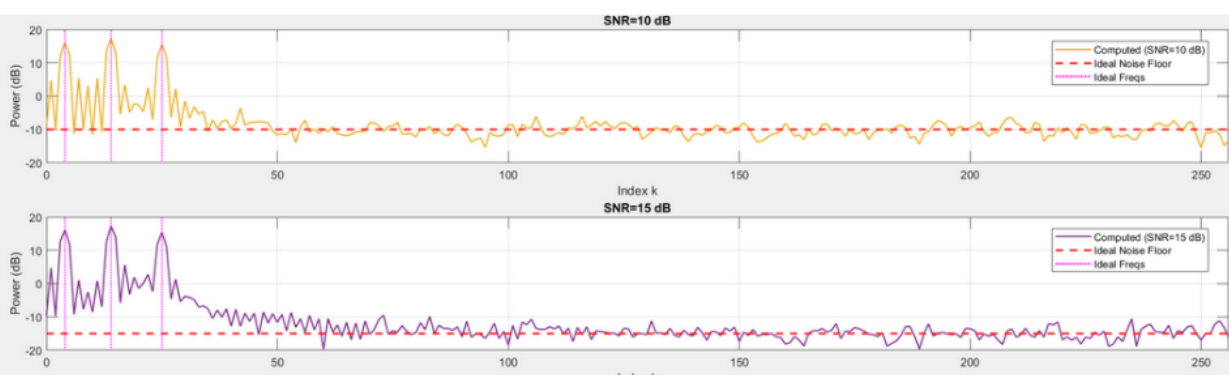
Q = 256, L = 256



Q = 1024, L = 256



Q = 2048, L = 256



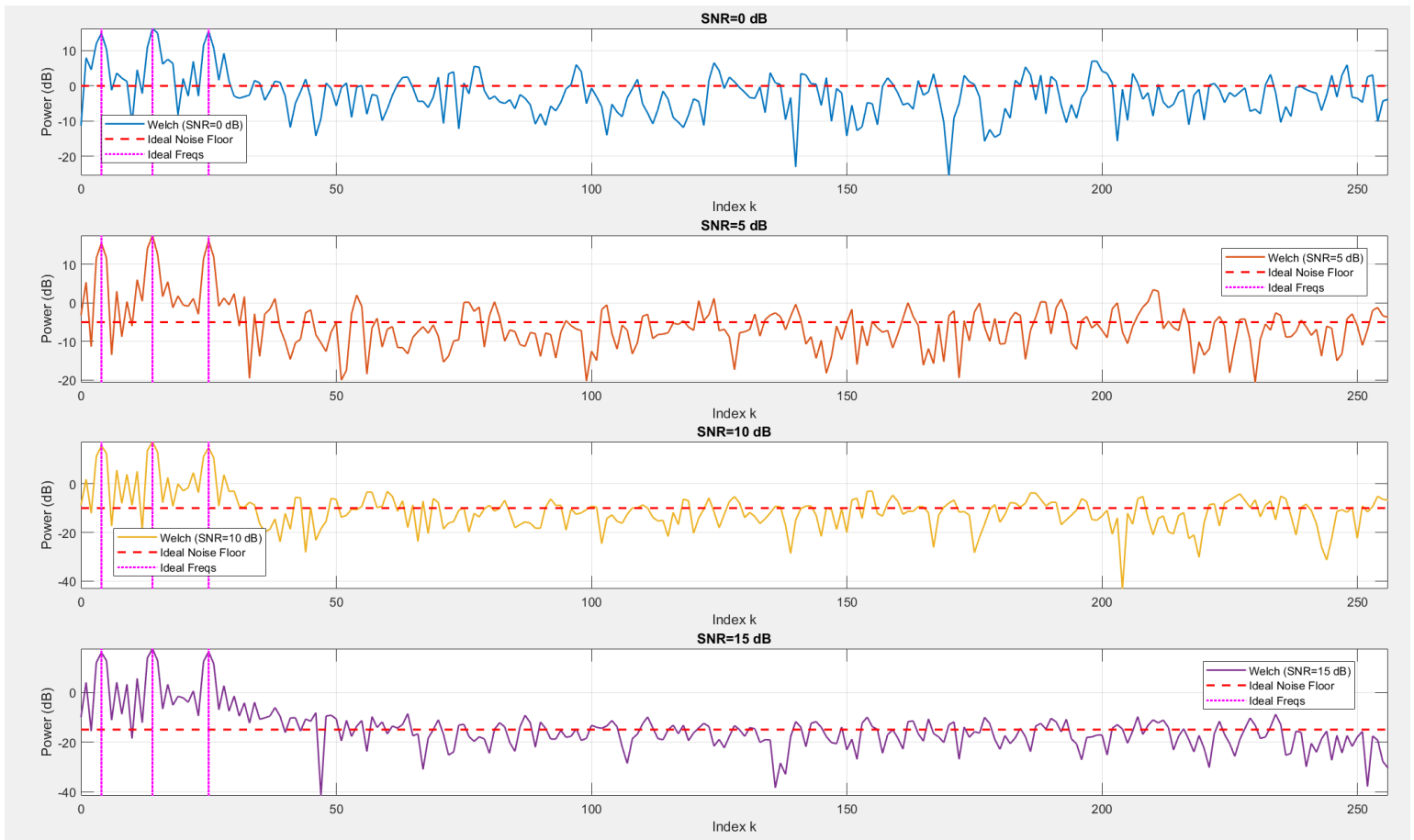
Examining the change in Q with all else held constant, we can see that the magnitude of the noise in the tail decreases as Q (the total number of sample points) is increased. The resolution of the peaks remains practically unchanged.

Welch Method Using the Rectangular Window

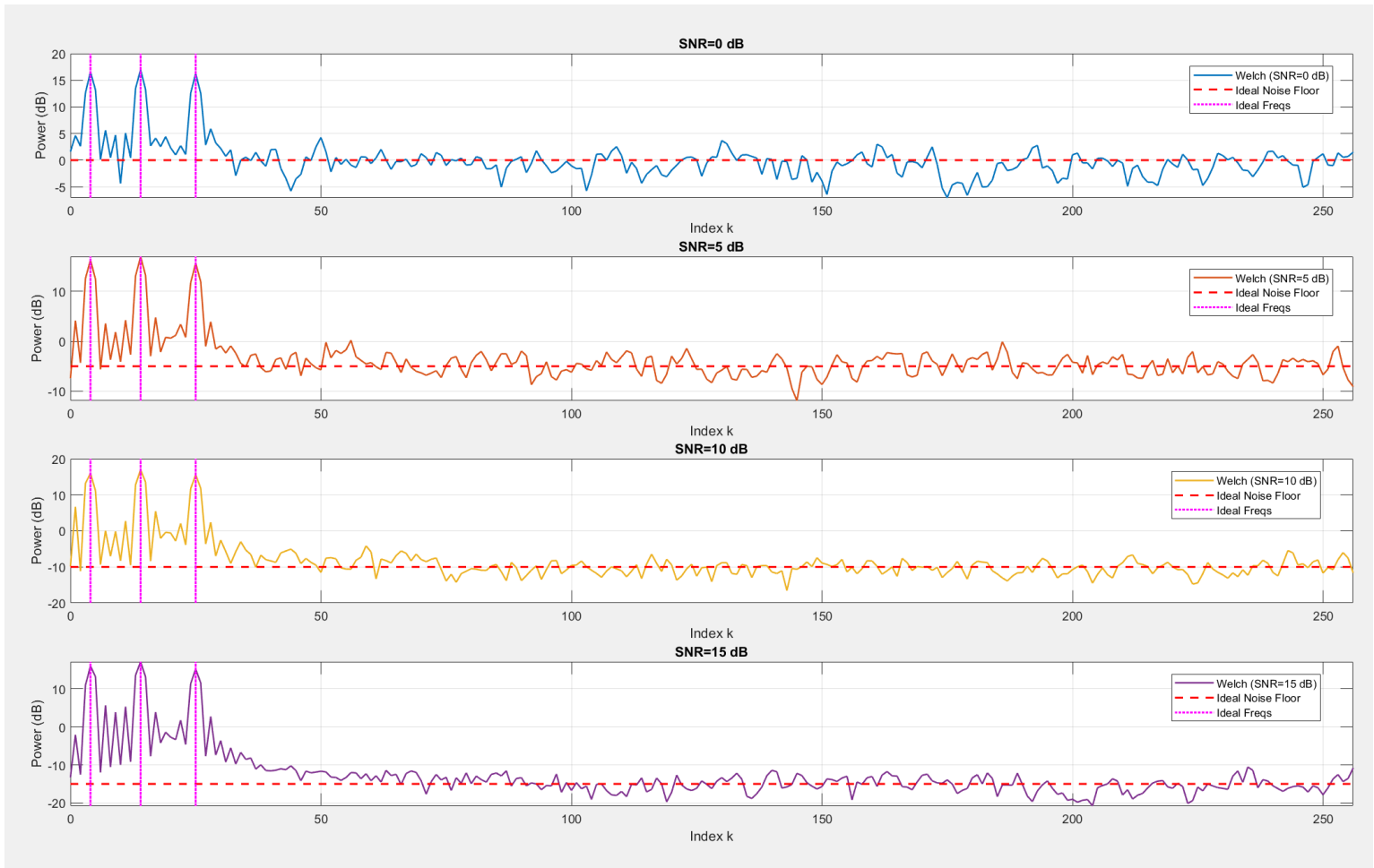
The Welch method differs from the Bartlett method in that it uses overlapping R and L values. This means that there will be a larger total number of segments, K, which provides the advantage of being able to increase the window length to increase resolution, without the tradeoff of increasing noise (since the number of segments K will be increased).

Some examples for fixed sets of parameters are given below:

Q=256, L=128



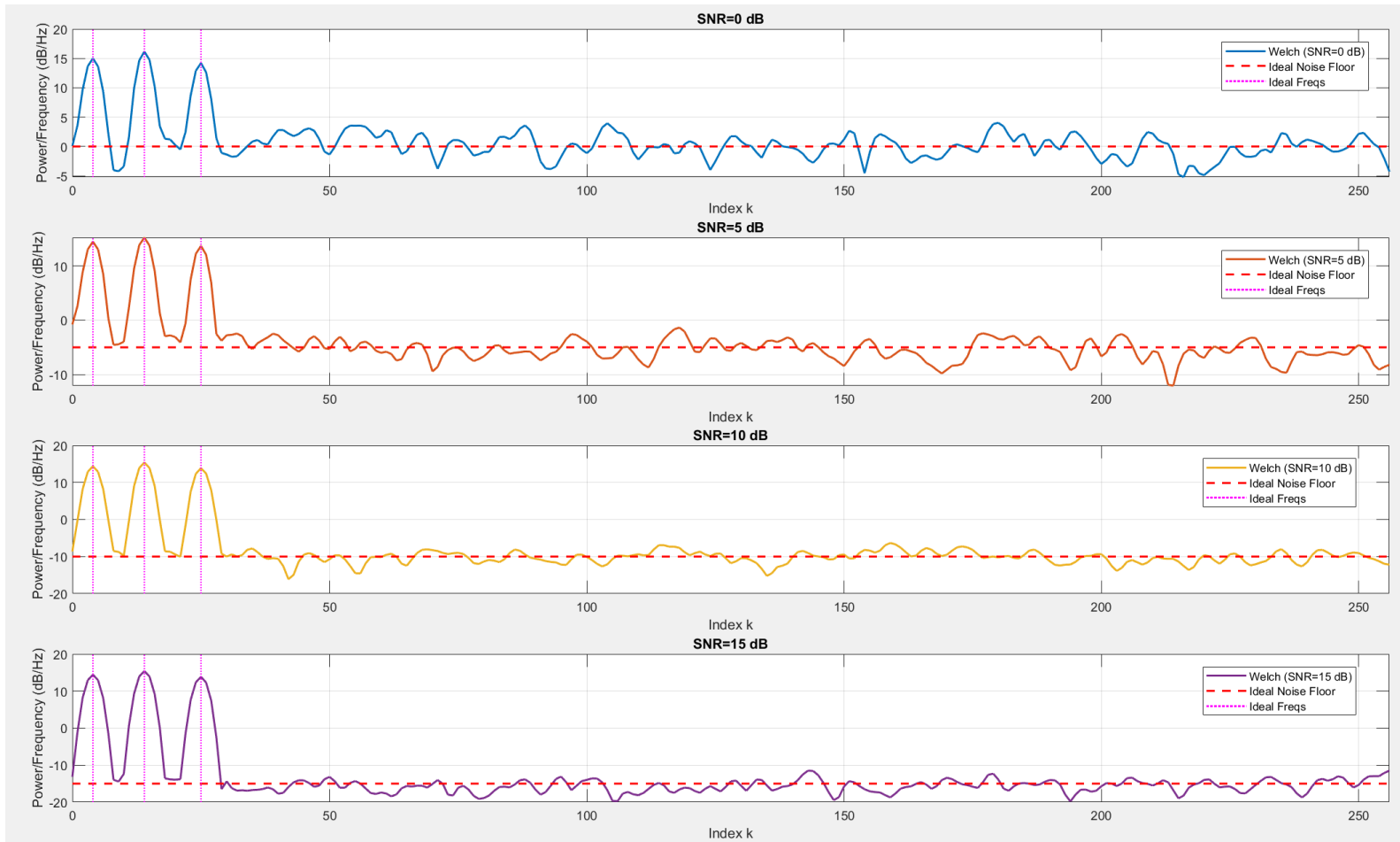
Q=1024, L=256



Welch Method Using the Kaiser Window

We again will use the Welch method, but this time changing the window used as well, from a rectangular window to a kaiser window. The beta and L value determine the Kaiser window and represent the trade off between main lobe width and side lobe attenuation. L is determined by trial and error, as shown above. The beta value helps control the side lobe attenuation, and a beta value of 6 gives approximately a side lobe attenuation of 50dB. Results from the average periodogram computation can be seen below:

Q = 1024 , L = 256



We can see that the noise is extremely low in the tail and the peaks, however, the resolution is not quite as sharp for the window function, although not much worse.

Comparing All Three Methods

From our analysis, we can see that to maximize our processing of the PSD, we want to do the following to our parameters:

- Maximize Q.
- Maximize K.
- Moderate value for L.
 - Want L high enough to have good resolution.

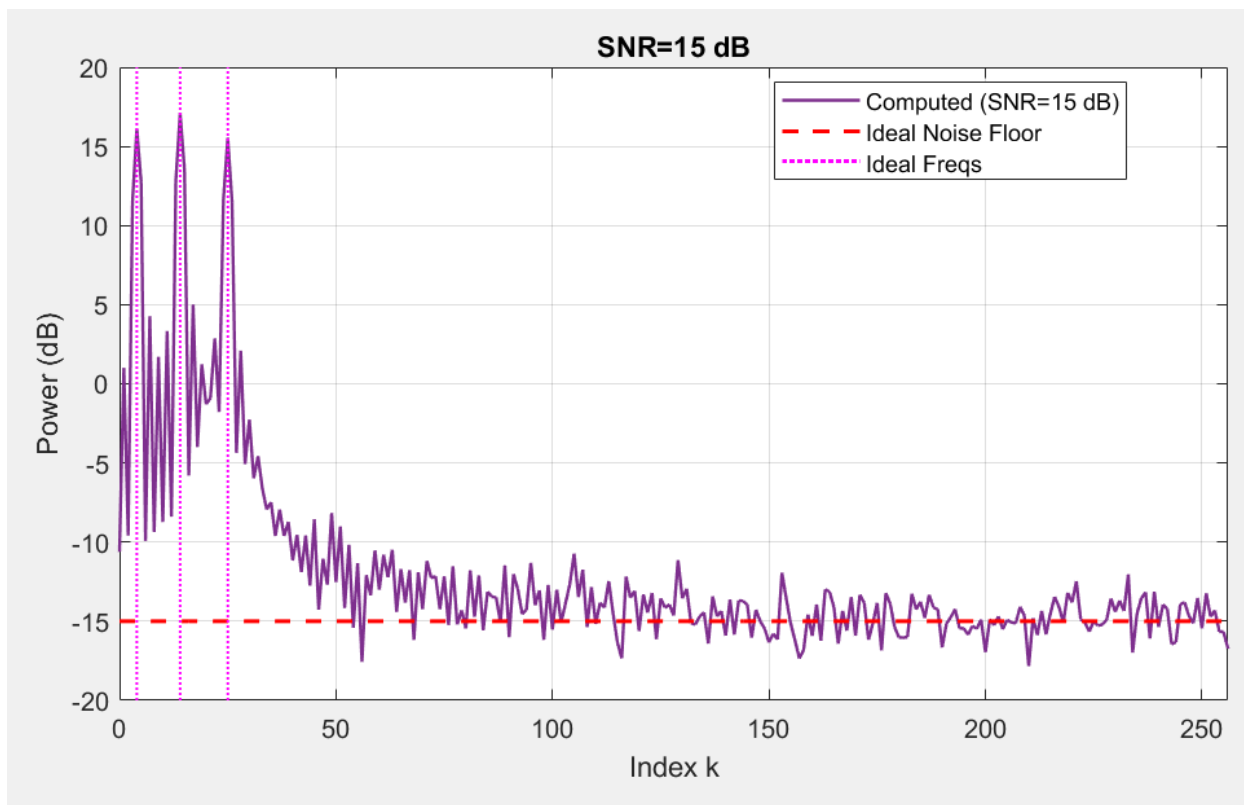
- Do not want L too high so that it reduces the noise tolerance by lowering K too much.
- Want R such that it maximizes the number of segments (K) but also does not create such small segments that we lose information while computing the average periodogram.
 - $R = L/2$ is good common practice.

Taking these notes into consideration, I will choose the following parameter values:

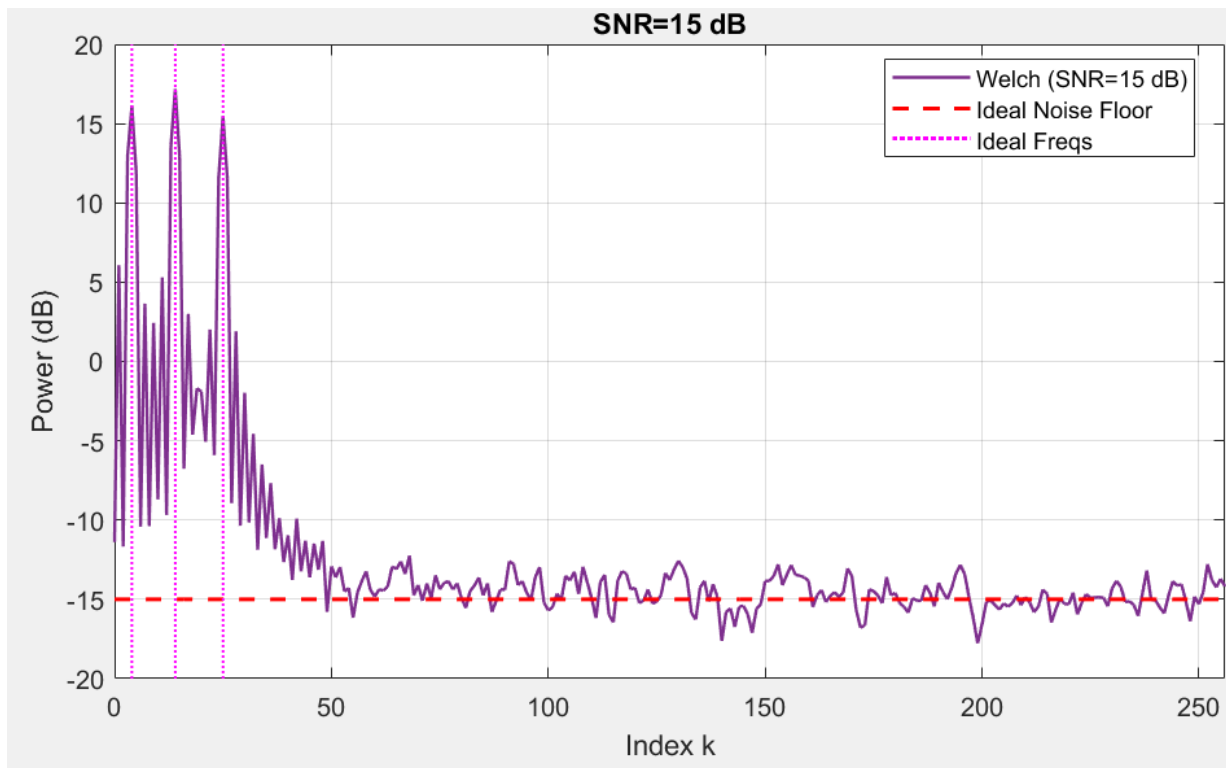
- $Q = 4096$
- $L = 256$
- $R = L$ (Bartlett), $L/2$ (Welch)
- $K = 16$ (Bartlett), 32 (Welch)

To compare the three methods, I will evaluate the computations based on their results for a SNR of 15 dB. The results can be seen below:

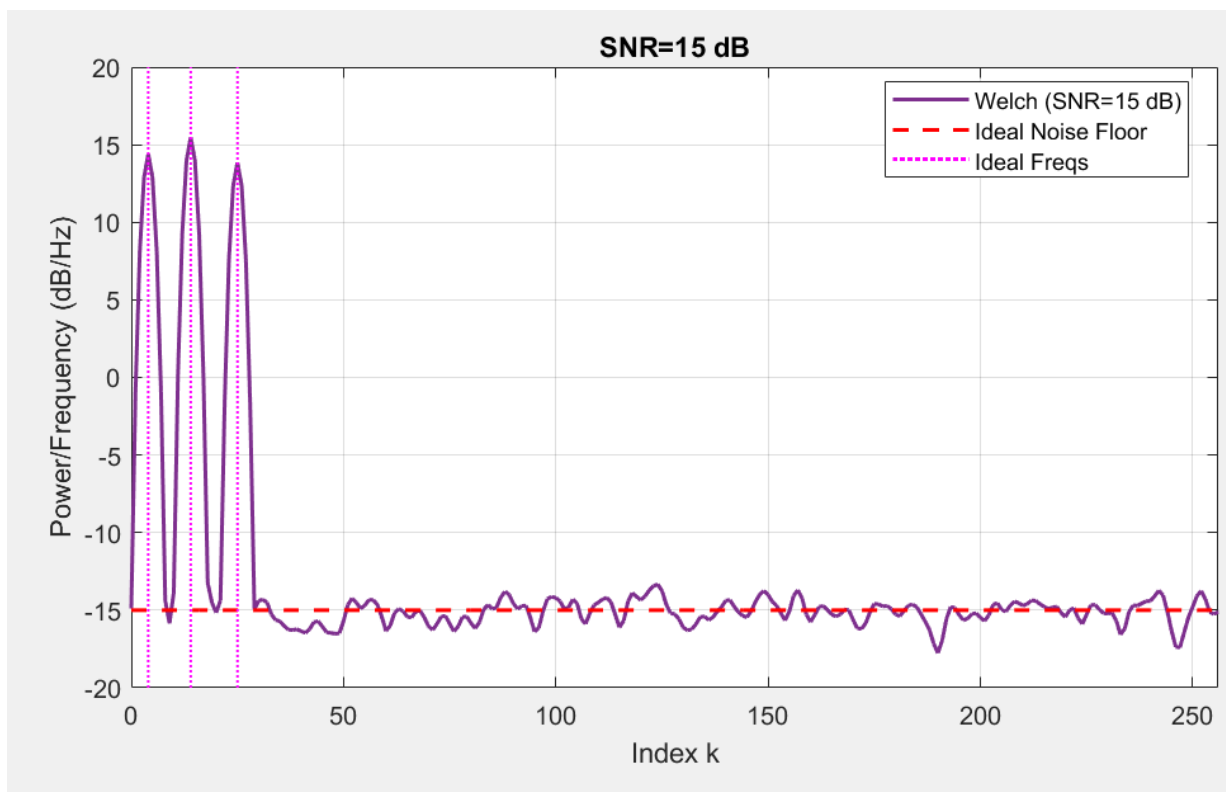
Bartlett



Welch, Rectangle Window



Welch, Kaiser Window



All of these results correctly compute the PSD and we can see that it matches exactly with the ideal overlay, even with 15 dB of white noise added to the signal. However, all three results are slightly different. Firstly, the Welch method results provide better attenuation of the noise in the tail of the signal's frequency response. Second, the windowing method has a significant impact on the final results. The rectangular window provides much sharper data that includes more of the noise. Meanwhile, the Kaiser window has a much smoother transition throughout the graph. At the peaks, the Kaiser window has slightly wider peaks, however, it does not include any of the spiky noise that is in between each of the three spikes. It also has the sharpest transition from the pass band to the stop band of all the graphs, making the peaks more apparent. Whichever method you choose it up to the user based on the situation, however, the notes outlining how to choose the values provide a clear guideline for how to configure your system.